Intergenerational Housing Misallocation and Fertility
Compact Analytical Model

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Abstract

Children take up space, and the space they take up is scarce in a particular way: family-sized units are concentrated in the owner-occupied segment of the housing market, while buying owner housing requires a down payment and debt-service capacity that many young households do not have. This note develops a compact stationary two-period OLG theory of that friction. Each period, young households decide whether to enter a housing market, choose between renting and owning, choose how many children to have, and save for old age; births, together with a renewal inflow, generate the next young cohort. Young owners become next period's old incumbents, who decide how much of their carried housing to keep before exiting. The model delivers three results. First, every constraint on family-sized space—rental size menus, down payments, payment-to-income limits—collapses into a single object, an effective family-housing cap, and a binding cap raises the price of a child through the space the child requires. Second, the stationary competitive equilibrium clears the housing market and still misallocates the existing stock whenever young households are priced out of owner housing by financing constraints while old incumbents are locked in place by capital-gains taxes that sale triggers and death forgives; the same tax, anticipated, raises the price of owning family-sized space for the young. Both gaps are measurable, one from household bundles and one from deeds and tax schedules. Third, at fixed prices, a housing policy changes aggregate fertility through constrained exposure, pass-through to the binding cap, the household fertility response, entry, and tenure switching.

The market structure follows Coven et al. (2025): tenure shapes which size menus and financing constraints a household faces, while all floorspace clears in a single market; supply margins are deferred to an extension. Children require housing space, so anything that rations family-sized space raises the private price of children. Family-sized space is rationed by tenure: by the size composition of the rental stock on one side, and by down-payment and payment-to-income requirements on the other. The old-side force—capital-gains taxes that fall due on sale but are forgiven at death—amplifies the scarcity but is not needed for any of the young-side results.

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Part I

Compact Analytical Model

1 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i, B_i),$$

where $y_i > 0$ is disposable nonhousing income, $a_i \geq 0$ is liquid wealth, and $B_i \geq 0$ is the gross bequest assigned at entry, taxed at rate τ^b on receipt. She decides whether to enter the housing market and, inside, chooses tenure, occupied housing, fertility, and savings. Old households, indexed by j as young households are by i, come in two groups. Old incumbents are last period's young owners: each enters old age with financial wealth, the house she bought when young, and a possible accrued capital gain; she chooses consumption, how much housing to retain, and how large an estate to leave, then exits. Old renters are last period's young renter entrants; they rent in old age, carry no housing and no gain, and exit. Nonentrants live outside the modeled market in both periods.

Masses. Time is discrete. At the start of period t a mass $M_t > 0$ of potential young households is drawn from a distribution $G_{Y,t}$ over types. Next period's potential young are the maturing children of the current cohort plus a renewal inflow of mass R_t; entry then determines how many of them participate in the housing market. The period-t measures of old incumbents and old renters are $F_{O,t}$ and $F_{R,t}$. Let $\bar{M}$ denote the stationary cohort mass. In a stationary equilibrium, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, and $F_{R,t} = F_R$ are reproduced by the transition laws below.

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of the consumption good and $\kappa > 0$ units of housing services, so with consumption spending c_i, occupied housing h_i, and fertility n_i, young preferences are

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \qquad \alpha, \vartheta > 0.$$

One more child means less space and less consumption left for the parents: children compete with their parents for the same house and the same budget. Rearranging spending on consumption and housing at user cost q shows the price of that competition,

$$(c_i - \chi n_i) + q\,(h_i - \kappa n_i) + (\chi + \kappa q)\,n_i = c_i + q h_i :$$

the full price of a child is its consumption requirement plus the market value of the space it occupies. When access to space is constrained, the space component carries a premium. A useful benchmark is equal scaling, $\chi = \kappa q/\alpha$, under which one child uses consumption and housing in the same proportions as its parents. Because $\vartheta \log n_i$ rules out $n_i = 0$, every entered household has positive completed fertility: the model studies the intensive margin, not childlessness.

Old preferences are

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \qquad \gamma > 0,$$

over consumption, retained housing, and the estate $e_j \geq 0$ the household leaves when she exits, valued through a warm-glow function $\mathcal{B}$, increasing and concave, possibly identically zero, and common to all old households. The exiting cohort's estates fund the bequests $B_{i'}$ received by the cohort that arrives, through the transition below. A young household also values her own old age: her objective adds $\beta\, u^O$, evaluated at the old-age choices she will make, with discount factor $\beta \in (0, 1]$; the old state she will hold is generated by the transition below.

Housing. There is a fixed aggregate stock $\bar{H}$ of divisible floorspace, measured in units of housing services. A household occupies floorspace in one of two tenure modes: renting, denoted R, or owning, denoted O. The modes offer different size menus,

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \qquad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \qquad 0 < \bar{h}^R < \bar{h}^O.$$

Family-sized houses are mostly available to buy, not to rent: a household that wants substantially more space than $\bar{h}^R$ must buy it.[1] All occupied floorspace trades in one market. The asset price of a unit of floorspace is P, and a no-arbitrage user-cost condition ties the rental rate to the price:

$$q = \rho P, \qquad \rho = r + \delta + \tau^p, \tag{1}$$

where r is the interest rate, δ is depreciation, and τ^p is a recurring property tax. Renters and owner-occupiers therefore face the same per-unit flow cost q on new occupancy choices; tenure matters through the size menus and through financing, not through the per-unit price of space, and the capital-gains tax below shifts only old incumbents' retention margin. Given q, a higher property tax lowers the asset price P and hence the down payment needed to buy a given amount of space. Throughout, lowercase h denotes an occupancy choice; capital H denotes housing quantities not chosen in the current period—the stock $\bar{H}$, effective caps, carried housing, aggregate demand. At the start of a stationary period, old incumbents carry into old age the houses generated by previous young owner choices, and passive landlords hold the residual stock:

$$H_{\text{pass}} = \bar{H} - \int H_j^0 \,\mathrm{d}F_O(j) \geq 0.$$

In a stationary equilibrium the inequality is implied by market clearing: the previous cohort's owner purchases cannot exceed the stock. Housing payments are transfers across households, landlords, and fiscal accounts, closed by lump-sum rebates, and reallocating the existing stock costs no real resources.[2]

Financing. Buying floorspace requires financing. A young buyer can finance at most a share $\phi \in (0, 1)$ of the purchase and faces a payment-to-income limit $\psi \in (0, 1)$ on housing outlays. Pledgeable resources for the down payment are

$$A_i(x) = a_i + xB_i + T_i^b(x), \qquad x = 1 - \tau^b,$$

where τ^b is an estate tax and $T_i^b(x)$ is the estate-tax rebate assigned to type i. The down payment is posted at purchase, at the start of the period, out of wealth in hand: income accrues over the period, so $A_i(x)$ caps what a buyer can put down. Wealth is otherwise fungible. It enters the young budget below alongside income and finances consumption and saving; buying assets is not expenditure, so

[1] Coven et al. (2025) model the same segmentation as a minimum size for owned homes, with rentals available in any size; we cap rental size instead. Both make family-sized space owner-concentrated; ours keeps the renter problem interior at the bottom.

[2] Two supply extensions are deliberately left out of the baseline and developed later: a price-elastic aggregate supply curve as in Coven et al. (2025), and tenure-specific stocks with separate construction margins, under which the rental and owner user costs diverge endogenously. The local results below survive with q replaced by the relevant tenure-specific user cost.

the posted equity remains part of the household's wealth. Inheritances and the estate tax therefore move both access to owner housing and lifetime resources. Households save: a young household carries $a_i' \geq 0$ into old age at the exogenous gross return $1+r$, and cannot borrow unsecured against old age. A renter, with no asset to pledge, therefore cannot borrow at all; a buyer's only debt is the financed share ϕ of the purchase. Newborn households inherit the estates of the old who exit as they arrive: B_i is assigned from old estates by the young-type transition below, and the estate tax falls on these transfers. The interest rate r is exogenous: households borrow and lend with a passive financial sector, so no domestic capital-market clearing condition is imposed, and the only price determined inside the model is q.

Capital-gains taxation. A housing sale during life realizes the seller's accrued capital gain, taxed at rate τ^g above an exclusion. Housing held until death passes to heirs with a stepped-up basis, and the gain is never taxed. For an old incumbent the difference between the two disposal routes is a per-unit tax $\ell_j \geq 0$ on housing sold while alive.

Where the gains come from is a modeling device, and worth stating plainly. The model is real and stationary; the tax code computes gains in dollars. Let all prices and incomes carry steady inflation $\varpi > 0$ with every relative price constant. A unit bought one period ago then shows the same nominal gain for every cohort—its real value never moved—and selling it triggers the real tax

$$\bar{\ell} = \tau^g \, \frac{P\,\varpi}{1+\varpi}$$

per unit: the inflation gain, taxed at sale, in today's goods. Inflation enters the model nowhere else; budgets, prices, and child requirements are real, and the exclusion is held fixed in real terms. So $\ell_j = \bar{\ell}$ for an old incumbent whose gain clears the exclusion and $\ell_j = 0$ below it, and because the exclusion is a level, who is locked in is decided by size: larger holdings clear it. The gain is deterministic, so a buyer knows the tax her future self will face; Section 7 prices that anticipation.[3]

Demography and transitions. The period timing is: potential young types are drawn; young households decide whether to enter; entrants choose tenure, housing, fertility, and savings; old incumbents choose consumption, retained housing, and estates; old households exit; births and the renewal inflow form the next potential-young cohort; current young owners become next period's old incumbents. Let

$$\Gamma_O(\mathrm{d}j' \mid i, h, a'; q, \tau^p)$$

be the old-incumbent transition for a current young owner of type i who buys h and carries savings a'. It assigns the house bought when young, $H^0_{j'} = h$; financial wealth

$$a_{j'} = (1+r)\,a' - q\,H^0_{j'},$$

so that total old wealth $a_{j'} + qH^0_{j'}$ is exactly the savings' return—the nominal gain is not real wealth; and the tax state, $\ell_{j'} = \bar{\ell}$ if the inflation gain on $H^0_{j'}$ clears the exclusion and zero otherwise. Housing wealth is in per-period goods units: the old live one more period, so a carried unit is worth one period of services q to its holder, and the shell that outlives her recirculates through the stock. The asset price P appears only in the down-payment requirement. Mortgage balances from youth are netted inside $a_{j'}$: a household that carried a mortgage has lower financial wealth and the same total. The draw does not condition on fertility: estates go to the arriving cohort, not to own children, so the model has no old-age-support motive for births. Young renter entrants map into

[3] Assessment protection of the Proposition 13 type, under which incumbents are taxed on below-market assessed values until they move, generates the same per-unit retention wedge.

old-renter states through a parallel kernel $\Gamma_R(\mathrm{d}j' \mid i, a')$, which assigns $a_{j'} = (1+r)\,a'$; nonentrants receive zero mass under both kernels. Old-renter states follow the same law as (2) with Γ_R and the renter indicator. Given current young choices, the next old-incumbent measure satisfies

$$F_{O,t+1}(A) = M_t \int \pi^E_{i,t} \mathbf{1}\{o_{i,t} = O\} \Gamma_O(A \mid i, h^O_{i,t}, a'^O_{i,t}; q_t, \tau^p_t)\, \mathrm{d}G_{Y,t}(i) \tag{2}$$

for every measurable set A of old states. In stationarity, this measure equals F_O.

Let

$$n^Y_{i,t} = \mathbf{1}\{o_{i,t} = R\} n^R_{i,t} + \mathbf{1}\{o_{i,t} = O\} n^O_{i,t} \tag{3}$$

denote fertility in the chosen inside branch, and let $\bar{n}^E_i \geq 0$ denote fertility in the outside option. The baseline formulas set $\bar{n}^E_i = 0$, but the demographic law allows outside fertility or inflow to renew the population.

Young type $i' = (y_{i'}, a_{i'}, B_{i'})$ is generated by fertility, old bequests, and the renewal inflow. Let

$$\Gamma_Y(\mathrm{d}i' \mid i, m, n, F_O, F_R, e; \Upsilon), \qquad m \in \{R, O, E\},$$

be the young-type kernel for a birth associated with current type i, branch m, fertility n, old bequest policy e, and renewal law Υ. The old who exit at the end of period t endow the cohort that arrives at $t+1$: the kernel assigns their bequest expenditure as the entrants' received bequests $B_{i'}$ and supplies the noninheritance components of young types. In stationarity, Γ_Y is restricted so that assigned bequests aggregate to the estates of all exiting old households, incumbents and renters:

$$\bar{M} \int \Big[\pi^E_i\, n^Y_i\, \mathbb{E}[B_{i'} \mid i, o_i] + (1 - \pi^E_i)\, \bar{n}^E_i\, \mathbb{E}[B_{i'} \mid i, E]\Big] \mathrm{d}G_Y(i) = \int e_j\, \mathrm{d}(F_O + F_R)\,(j),$$

where the expectations are over Γ_Y and renewal draws from Υ receive no bequests. Let $R_t \geq 0$ be the renewal inflow, outside births or immigration, with type law Υ. The next young cohort satisfies, for every measurable set A of young types,

$$\begin{aligned} M_{t+1} G_{Y,t+1}(A) = M_t \int \Big[&\pi^E_{i,t} n^Y_{i,t} \Gamma_Y(A \mid i, o_{i,t}, n^Y_{i,t}, F_{O,t}, F_{R,t}, e_t; \Upsilon) \\ &+ (1 - \pi^E_{i,t}) \bar{n}^E_i \Gamma_Y(A \mid i, E, \bar{n}^E_i, F_{O,t}, F_{R,t}, e_t; \Upsilon)\Big] \mathrm{d}G_{Y,t}(i) + R_t \Upsilon(A). \end{aligned} \tag{4}$$

Setting A equal to the whole young type space gives the cohort-mass law:

$$M_{t+1} = M_t \int \big[\pi^E_{i,t} n^Y_{i,t} + (1 - \pi^E_{i,t}) \bar{n}^E_i\big]\, \mathrm{d}G_{Y,t}(i) + R_t. \tag{5}$$

In a stationary equilibrium, $M_t = \bar{M}$, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, $F_{R,t} = F_R$, and $R_t = \bar{R}$. If $\bar{R} = 0$, stationarity requires average births per potential young household to be one. If $\bar{R} > 0$, it requires average births below one, with the shortfall covered by renewal.

Entry. A potential young household can enter the housing market or take an outside option with lifetime value $\bar{W}^E$; entry is subject to Type-I extreme-value taste shocks with scale $\kappa_E > 0$.

2 The young household's problem

Conditional on entry, household i takes the user cost q and the asset price $P = q/\rho$ as given and chooses a tenure mode $m \in \{R, O\}$, a bundle, and savings. The two modes share one objective and one budget; they differ in the size menu and in whether the financing constraints apply. For $m \in \{R, O\}$,

$$W_i^m(q, \tau^p, x) = \max_{c_i, h_i, n_i, a_i'} \Big\{ \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ + \beta \big[\mathbf{1}\{m = R\}\, V^R(a_{j'}) + \mathbf{1}\{m = O\}\, V^O(a_{j'}, h_i, \ell_{i'}) \big] \Big\} \tag{6}$$

subject to

$$c_i + q h_i + a_i' = y_i + A_i(x), \tag{7}$$
$$h_i \in \mathcal{H}^m, \qquad a_i' \geq 0,$$
$$a_{j'} = (1 + r)\, a_i' - \mathbf{1}\{m = O\}\, q h_i, \tag{8}$$
$$\mathbf{1}\{m = O\}\, (1 - \phi) P h_i \leq A_i(x), \tag{9}$$
$$\mathbf{1}\{m = O\}\, q h_i \leq \psi y_i, \tag{10}$$
$$c_i - \chi n_i > 0, \qquad h_i - \kappa n_i > 0.$$

V^O and V^R are the old-age values defined in the next section, evaluated at the state the household generates; $\ell_{i'} = \bar{\ell}$ if h_i clears the exclusion and zero below. Nothing here is random: the household knows the old state her choices produce. The budget is the same in both modes—space costs q per unit either way, wealth is spendable either way, savings earn $1 + r$ either way—so tenure matters through the menu and the two owner financing requirements, not through prices.

Using $P = q/\rho$, owner housing is bounded above by

$$h_i \leq H_i^O(q, \tau^p, x) = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho}{(1 - \phi) q}, \frac{\psi y_i}{q} \right\}.$$

The renter analog is $H_i^R = \bar{h}^R$. In mode m, H_i^m is the effective family-housing cap. A renter is capped by the rental menu. A would-be buyer is capped by whichever of the owner menu, down-payment capacity, or payment-to-income capacity is shortest. A policy affects fertility through this channel only when it relaxes the binding component of H_i^m. If the owner branch has no feasible allocation, for example when $A_i(x) = 0$, set $W_i^O = -\infty$.

The inside value of entry is

$$W_i^Y(q, \tau^p, x) = \max\{W_i^R(q), W_i^O(q, \tau^p, x)\}. \tag{11}$$

Let $o_i \in \{R, O\}$ denote an optimal tenure.[4]

3 The old household's problem

The baseline young-access mechanism below does not require any old-side force. When the old side is active, the compact model uses the capital-gains institution from the environment. On old-household objects (V^O, V^R, u^O, c_j^O) the superscripts mark the old stage and its tenure; the

[4] The hard rental cap $\bar{h}^R$ is the simplest representation of size segmentation. A smooth alternative lets renters pay $R(h) = qh + \varrho(h)$ with $\varrho_h(h) \geq 0$ for large units, so that large rentals exist but at a premium; every marginal condition below goes through with q replaced by $R_h(h)$. Such a premium is also one natural source of a rent–own gap in per-unit costs, which Section 9 studies.

index j separates old households from young ones. Old incumbent j is last period's young owner and enters with financial wealth a_j (negative when a mortgage balance remains), carried housing H^0_j, and an accrued gain on it. Each unit she holds faces one choice: sell it now and pay the gains tax, or keep it until death, when the basis steps up and the tax is forgiven. ℓ_j is the tax per unit that selling triggers: $\bar{\ell}$ when her gain clears the exclusion, zero below.[5] She chooses consumption, retained housing, and her estate, with the preferences of Section 1:

$$\begin{aligned} V^O(a_j, H^0_j, \ell_j) = \max_{c^O_j, h^O_j, e_j} \;& u^O(c^O_j, h^O_j, e_j) \\ \text{subject to} \quad & c^O_j + e_j + q\, h^O_j = a_j + qH^0_j - \ell_j(H^0_j - h^O_j), \qquad 0 \le h^O_j \le H^0_j. \end{aligned} \tag{12}$$

Releasing a unit of housing nets $q - \ell_j$, its market value less the tax the sale realizes, while the market values the released space at q. The forgiveness at death is a subsidy to staying put: a tax rule attached to the disposal route, not a preference and not a real resource saving. The planner below counts true old housing utility and true warm-glow bequest utility but does not count the avoided tax as a social benefit. An old household that keeps a large house because she values it is using housing efficiently; an old household that keeps it because selling would realize a taxable gain is not.

An old renter carries no housing and no tax: her wealth is her returned savings, $a_j = (1 + r)a'_i$, positive because zero savings would leave her nothing to consume. With the same preferences she solves

$$V^R(a_j) = \max_{c^O_j, h^O_j, e_j} u^O(c^O_j, h^O_j, e_j) \quad \text{subject to} \quad c^O_j + e_j + q\, h^O_j = a_j, \qquad 0 < h^O_j \le \bar{h}^R, \tag{13}$$

and at an interior choice $\gamma c^O_j / h^O_j = q$: old rental demand adds to market clearing but carries no wedge.[6]

4 Entry

With the entry technology of Section 1, the probability that type i enters the housing market is

$$\pi^E_i(q, \tau^p, x) = \frac{\exp\{W^Y_i(q, \tau^p, x)/\kappa_E\}}{\exp\{W^Y_i(q, \tau^p, x)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \tag{14}$$

The endogenous measure of young entrants is

$$\mathrm{d}F^E_Y(i; q, \tau^p, x) = \bar{M}\, \pi^E_i(q, \tau^p, x)\, \mathrm{d}G_Y(i).$$

The limit $\kappa_E \downarrow 0$ gives deterministic entry, with the indifference set $W^Y_i = \bar{W}^E$ assumed to carry zero mass. Conditional on entry, the household chooses tenure according to (11).

[5] The statutory tax is triggered by the sale of the discrete dwelling; with divisible floorspace the per-unit tax is its smooth counterpart. Discreteness concentrates the same wedge at the downsizing margin and strengthens retention.

[6] Coven et al. (2025) model this lock-in channel in lifecycle form and evaluate a rate reduction. The tax ℓ_j prices the same margin statically, and also prices the opposite reform: taxing accrued gains at every transfer, including bequest, removes the step-up and shrinks ℓ_j to the value of tax deferral.

5 Competitive equilibrium

A stationary equilibrium clears one aggregate floorspace market and closes the old-incumbent, young-type, and cohort-mass fixed points. Households take the user cost as given, choose tenure and housing, and their choices aggregate into demand for the fixed stock. There is no pairwise assignment of old units to young households: one price equates total demand to the stock, and nothing in that condition asks whether the marginal family-sized unit is held by the household that values it most. That question is deferred to the efficiency analysis below.

Given policies (τ^p, x), the stock $\bar{H}$, the stationary cohort mass $\bar{M}$, young distribution G_Y, old measures F_O and F_R, and transfer rules, aggregate floorspace demand $H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R)$ at a candidate user cost q adds young households, old incumbents, and old renters:

$$H^D = \bar{M} \int \pi_i^E \left[\mathbf{1}\{o_i = R\} h_i^R + \mathbf{1}\{o_i = O\} h_i^O\right] \mathrm{d}G_Y(i) + \int h_j^O \, \mathrm{d}F_O(j) + \int h_j^O \, \mathrm{d}F_R(j).$$

Tenure determines which menu and which financing constraints apply to a household, not which market it buys space in: all of this demand falls on the same stock.

Definition 1 (Stationary competitive equilibrium). *Given $(\tau^p, x, \bar{H})$, transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, and a stationary renewal law $(\bar{R}, \Upsilon)$, a stationary competitive equilibrium consists of a stationary cohort mass $\bar{M} > 0$, a user cost q, an asset price P, a stationary young distribution G_Y, a stationary old-incumbent measure F_O, a stationary old-renter measure F_R, young branch allocations including savings, tenure choices, entry probabilities, old allocations, and fiscal transfers such that:*

1. *conditional on entry, young households solve (6) in each mode, then choose tenure by (11);*
2. *old incumbents solve (12) and old renters solve (13);*
3. *entry is given by (14);*
4. *the asset price satisfies the no-arbitrage condition (1);*
5. *the single floorspace market clears,*
$$H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R) = \bar{H}; \tag{15}$$
6. *passive housing-income, credit, property-tax, capital-gains, estate-tax, and policy-instrument accounts are closed by lump-sum transfers;*
7. *the old-incumbent measure is stationary: for every measurable set A of old states,*
$$F_O(A) = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} \Gamma_O(A \mid i, h_i^O, a_i'^O; q, \tau^p) \, \mathrm{d}G_Y(i), \tag{16}$$
and the old-renter measure satisfies the same condition with Γ_R and $\mathbf{1}\{o_i = R\}$;
8. *the young cohort is stationary: for every measurable set A of young types,*
$$\begin{aligned} \bar{M} G_Y(A) = \bar{M} \int \Big[& \pi_i^E n_i^Y \Gamma_Y(A \mid i, o_i, n_i^Y, F_O, F_R, e; \Upsilon) \\ & + (1 - \pi_i^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_O, F_R, e; \Upsilon) \Big] \mathrm{d}G_Y(i) + \bar{R}\Upsilon(A), \end{aligned} \tag{17}$$
where e is the old bequest-expenditure policy. Taking A to be the whole young type space gives the stationary cohort-balance condition
$$\bar{M} = \bar{M} \int \left[\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E\right] \mathrm{d}G_Y(i) + \bar{R}. \tag{18}$$

Let $\Xi^{\Pi} \geq 0$ denote real resource costs of policy instruments. The transfer rule is chosen so that, after summing household budgets and using (15), the aggregate goods constraint is

$$\bar{M} \int \pi_i^E \left[\mathbf{1}\{o_i = R\}\big(c_i^R + a_i'^R - y_i - A_i(x)\big) + \mathbf{1}\{o_i = O\}\big(c_i^O + a_i'^O - y_i - A_i(x)\big) \right] \mathrm{d}G_Y(i)$$
$$+ \int (c_j^O + e_j - a_j)\, \mathrm{d}(F_O + F_R)\,(j) + \Xi^{\Pi} \leq 0.$$

Housing quantities are disciplined separately by floorspace feasibility: housing payments are transfers that net out against rebates, and the old carried-housing term qH_j^0 is a claim on the existing stock rather than new output. The intergenerational books close through the transitions: bequest receipts net against old estates by the adding-up restriction on Γ_Y, estate-tax collections net against the rebates T_i^b, and the net return r on carried savings is paid by the passive financial sector, the open-economy counterpart of foreign asset income. Housing in this baseline is a pure allocation problem.

Aggregate births generated by the potential-young cohort are

$$N = \bar{M} \int \left[\pi_i^E n_i^Y + (1 - \pi_i^E)\bar{n}_i^E \right] \mathrm{d}G_Y(i).$$

In a stationary demographic equilibrium, $N + \bar{R} = \bar{M}$. Under the baseline outside-option normalization $\bar{n}_i^E = 0$, $N = \bar{M} \int \pi_i^E n_i^Y \, \mathrm{d}G_Y(i)$.

6 Constrained planner

The planner comparison is evaluated at a stationary competitive equilibrium. For the local variations below, the stationary cohort mass $\bar{M}$, young distribution G_Y, old measures F_O and F_R, transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, renewal inflow $\bar{R}$, and tax rules are held fixed at the instant of the variation. The question is which constraints young families face as real scarcity and which are financing arrangements that a different institutional design could undo.

The planner has the same preferences, stationary young distribution, old measures F_O and F_R, entry technology, housing stock, and tenure size menus as the competitive economy. It chooses entry-relevant inside allocations, tenure assignments, young consumption, housing and fertility, old consumption, housing and estates, and policy instruments for the current stationary cross-section. It does not move the savings margin: young savings a_i', and the old wealth they generate, are held at their equilibrium values, so the comparison is within-period reallocation given the intergenerational wealth transition. The planner respects floorspace feasibility and the rental and owner menus.

The planner faces the economy's real constraints and nothing else: the floorspace stock, the tenure size menus, goods feasibility, and the domain conditions below. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy; when an allocation must be implemented through market purchases, Section 8 prices the required instruments by their marginal real cost. The planner counts true old housing utility and true warm-glow bequest utility; it does not count the avoided gains tax ℓ_j, which is a transfer.

Let $\omega_i^Y > 0$ and $\omega_j^O > 0$ be Pareto weights. Let $\mathcal{G}_i(n_i)$ denote an optional social payoff from births beyond parental utility. The fertility-neutral planner has $\mathcal{G}_i \equiv 0$.

If the planner assigns young type i tenure $o_i^P \in \{R, O\}$ and allocation (c_i, h_i, n_i), the inside value relevant for entry is

$$V_i^P = u_i^Y(c_i, h_i, n_i).$$

The induced entry probability is

$$\pi_i^P = \frac{\exp\{V_i^P/\kappa_E\}}{\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}.$$

The corresponding expected entry value, up to an additive constant, is

$$\mathcal{W}_i^E(V_i^P) = \kappa_E \log\left[\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}\right].$$

The planner solves

$$\max \bar{M} \int \omega_i^Y \mathcal{W}_i^E(V_i^P)\, \mathrm{d}G_Y(i) + \bar{M} \int \pi_i^P \mathcal{G}_i(n_i)\, \mathrm{d}G_Y(i) + \int \omega_j^O u^O(c_j^O, h_j^O, e_j)\, \mathrm{d}(F_O + F_R)(j) \tag{19}$$

subject to floorspace feasibility,

$$\bar{M} \int \pi_i^P \left[\mathbf{1}\{o_i^P = R\} + \mathbf{1}\{o_i^P = O\}\right] h_i \, \mathrm{d}G_Y(i) + \int h_j^O \, \mathrm{d}(F_O + F_R)(j) \le \bar{H},$$

menu feasibility,

$$h_i \in \mathcal{H}^R \text{ if } o_i^P = R, \qquad h_i \in \mathcal{H}^O \text{ if } o_i^P = O,$$

and goods feasibility,

$$\begin{aligned} \bar{M} \int \pi_i^P \big(c_i + a_i' - y_i - A_i(x)\big) \mathrm{d}G_Y(i) & \\ + \int (c_j^O + e_j - a_j)\, \mathrm{d}(F_O + F_R)(j) + \Xi^\Pi \le 0, & \end{aligned}$$

All allocations also satisfy

$$c_i - \chi n_i > 0, \qquad n_i > 0, \qquad h_i - \kappa n_i > 0,$$

for young households, and

$$c_j^O > 0, \qquad e_j \ge 0, \qquad 0 \le h_j^O \le H_j^0,$$

for old incumbents, with the retention bound replaced by the rental menu for old renters.

Let Λ be the multiplier on goods feasibility and ΛQ the multiplier on floorspace feasibility, so that Q is the planner's scarcity price of space in consumption goods. For a fixed tenure assignment and active set, and away from menu corners, an interior planner allocation satisfies

$$\begin{aligned} \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q && \text{for young households,} \\ \frac{u_{h,j}^O}{u_{c,j}^O} &= Q && \text{for old housing,} \end{aligned}$$

$$\frac{\vartheta\,(c_i-\chi n_i)}{n_i}+\nu_i^g=\chi+\kappa Q,$$
$$\frac{\mathcal{B}'(e_j)}{u^O_{c,j}}=1 \qquad \text{if } e_j>0.$$

The planner equalizes every household's housing-consumption tradeoff to the common scarcity price of space, prices children at $\chi+\kappa Q$, and respects the bequest motive. The term ν_i^g is the goods-equivalent social value of a marginal birth beyond parental utility. If $\mathcal{G}_i\equiv 0$, then $\nu_i^g=0$. Endogenous logit entry affects the levels of the first-order conditions through a common factor, but that factor cancels from the housing-consumption and fertility ratios displayed here.

7 Marginal conditions in equilibrium

How tightly a constraint binds can be read off the household's own bundle. A household that is free to adjust equates her marginal value of housing space to the price of space. A household pinned by a size menu or a financing requirement values space above its price, and the distance between the two is the constraint's bite, measured in consumption goods.

Every condition below is a derivative of one Lagrangian. For mode m,

$$\begin{aligned}\mathcal{L}_i^m =\;& \log(c_i-\chi n_i)+\alpha\log(h_i-\kappa n_i)+\vartheta\log n_i+\beta\,V_{j'}\\ &+\lambda_i^m\big(y_i+A_i(x)-c_i-qh_i-a_i'\big)+\theta_i^m\big(\bar h^m-h_i\big)\\ &+\mathbf{1}\{m=O\}\Big[\mu_i\big(A_i(x)-(1-\phi)Ph_i\big)+\eta_i\big(\psi y_i-qh_i\big)\Big],\end{aligned}$$

with λ_i^m on the budget (7), $\theta_i^m\geq 0$ on the size menu, and $\mu_i\geq 0$ and $\eta_i\geq 0$ on the down-payment (9) and payment-to-income (10) constraints, active only for buyers. The generated old state (8) is substituted into $V_{j'}$; the savings bound $a_i'\geq 0$ appears as the complementarity attached to the Euler equation (24) below, and the domain conditions hold strictly at any optimum.

For a renter ($m=R$), at an interior allocation with respect to consumption and fertility,

$$\begin{aligned}\frac{1}{c_i^R-\chi n_i^R}&=\lambda_i^R,\\ \frac{\alpha}{h_i^R-\kappa n_i^R}&=\lambda_i^R q+\theta_i^R,\\ \frac{\vartheta}{n_i^R}&=\frac{\chi}{c_i^R-\chi n_i^R}+\frac{\alpha\kappa}{h_i^R-\kappa n_i^R}.\end{aligned}\tag{20}$$

Converting the cap multiplier into consumption goods, $\zeta_i^R=\theta_i^R/\lambda_i^R\geq 0$, the renter's marginal value of housing is

$$\frac{u^Y_{h,i}}{u^Y_{c,i}}=\frac{\alpha\,(c_i^R-\chi n_i^R)}{h_i^R-\kappa n_i^R}=q+\zeta_i^R.\tag{21}$$

A renter against the size cap values one more unit of space above the rent it would command, and the gap is computable from the observed bundle as $\zeta_i^R=\alpha(c_i^R-\chi n_i^R)/(h_i^R-\kappa n_i^R)-q$, the household's tradeoff between space and consumption, net of children, in excess of the market rent. An unconstrained renter has $\zeta_i^R=0$.

Assumption 1 (Regularity at the old margin)**.** *(i) The household's induced old retention choice is interior,* $h^O_{j'} < H^0_{j'}$*. (ii) Types at the size threshold where the inflation gain meets the exclusion have zero mass.*

Anticipation is a price here, not a friction to assume away. Buying one more unit raises anticipated old wealth by its housing value and costs that value today; above the exclusion it also commits the household to the tax $\bar{\ell}$ on the marginal unit her old self sells. By the envelope condition $\partial V^O/\partial H^0 = (q-\ell)/c^O$, which part (i) keeps smooth, the continuation contributes $-\beta\bar{\ell}/c^O_{j'}$ to the buyer's housing condition, and the Euler equation converts it into goods:

$$\frac{\beta\,\bar{\ell}\,(c_i-\chi n_i)}{c^O_{j'}} = \frac{\bar{\ell}}{1+r}.$$

A buyer above the exclusion therefore pays an effective price

$$q^O \equiv q + \frac{\ell_{i'}}{1+r}, \qquad \ell_{i'} \in \{0, \bar{\ell}\},$$

per unit of space—the market price plus her own discounted lock-in tax—while a renter, or a buyer below the exclusion, pays q. The tax acts twice: ex post it locks in the current old, and ex ante it taxes ownership of family-sized space.

For a young owner ($m = O$) above the exclusion, at an interior allocation with respect to consumption and fertility,

$$\begin{aligned}
\frac{1}{c^O_i - \chi n^O_i} &= \lambda^O_i, \\
\frac{\alpha}{h^O_i - \kappa n^O_i} &= \lambda^O_i q + \frac{\beta\,\bar{\ell}}{c^O_{j'}} + \mu_i(1-\phi)P + \eta_i q + \theta^O_i, \\
\frac{\vartheta}{n^O_i} &= \frac{\chi}{c^O_i - \chi n^O_i} + \frac{\alpha\kappa}{h^O_i - \kappa n^O_i}.
\end{aligned} \tag{22}$$

Below the exclusion the anticipated-tax term is absent. The same conversion for the owner collects the access constraints, each priced by the primitive that generates it:

$$\zeta^O_i = \underbrace{\frac{\mu_i}{\lambda^O_i}(1-\phi)P + \frac{\eta_i}{\lambda^O_i}q}_{\zeta^{O,F}_i,\ \text{financial component}} + \underbrace{\frac{\theta^O_i}{\lambda^O_i}}_{\text{owner-menu component}} \geq 0.$$

The first term is the collateral margin: it is positive when the household's pledgeable wealth $A_i(x)$ cannot cover the down payment $(1-\phi)P$ on a larger unit. The second is the debt-service margin: positive when the payment limit ψy_i binds. The third is the owner size menu. If the owner menu cap does not bind, then $\theta^O_i = 0$ and $\zeta^O_i = \zeta^{O,F}_i$: over and above her effective price, the household is constrained by finance, not by the stock. The young owner's marginal value of owner housing is

$$\frac{u^Y_{h,i}}{u^Y_{c,i}} = \frac{\alpha\,(c^O_i - \chi n^O_i)}{h^O_i - \kappa n^O_i} = q^O + \zeta^O_i. \tag{23}$$

Both tenure modes share the intertemporal condition

$$\frac{1}{c_i - \chi n_i} \;\geq\; \beta(1+r)\,\frac{1}{c^O_{j'}}, \qquad \text{with equality if } a'_i > 0, \tag{24}$$

the Euler equation, with $c^O_{j'}$ old consumption at the generated state: both old problems have envelope $\partial V/\partial a_j = 1/c^O_j$, and carried savings pay $1+r$. In the benchmark the Euler equation pins savings in closed form. With logarithmic warm glow, $\mathcal{B}(e) = b\log e$ with $b \geq 0$, old consumption at interior choices is proportional to liquidation wealth, and the Euler equation collapses to the savings rule

$$a'_i = k\,(c_i - \chi n_i) + \frac{\ell_{i'}\, h_i}{1+r}, \qquad k = \beta\,(1+\gamma+b)\,, \tag{25}$$

with $\ell_{i'} = 0$ for renters and below-exclusion buyers. In the first term the gross return drops out, the usual log cancellation; the second is the discounted future tax bill, which the household saves in advance: anticipating the lock-in tax is itself a savings motive. The savings load k is old age's claim on each unit of adult consumption; the comparative statics of Section 9 use this rule.

For an old owner at an interior choice of current housing to keep,

$$\frac{u^O_{h,j}}{u^O_{c,j}} = \frac{\gamma c^O_j}{h^O_j} = q - \ell_j. \tag{26}$$

The old owner's valuation sits *below* the market price by exactly the tax: she keeps marginal space she values at less than the market would pay, because selling it would realize the taxable gain. If bequests are interior, then

$$\mathcal{B}'(e_j) = \frac{1}{c^O_j},$$

which matches the planner's bequest condition: leaving wealth to children is a preference the model respects, not a distortion. An old renter at an interior choice has $u^O_{h,j}/u^O_{c,j} = q$ exactly; her demand is wedge-free.

Proposition 1 (Constrained family-housing access and fertility)**.** *Fix an entered young household and its optimal tenure branch. If the household rents, then*

$$\frac{\vartheta\,(c^R_i - \chi n^R_i)}{n^R_i} = \chi + \kappa(q + \zeta^R_i).$$

If the household owns, then

$$\frac{\vartheta\,(c^O_i - \chi n^O_i)}{n^O_i} = \chi + \kappa(q^O + \zeta^O_i),$$

with $q^O = q + \ell_{i'}/(1+r)$ the buyer's effective price. Thus rental size limits and owner access constraints enter fertility through the housing services required per child.

Proof. For renters, multiply (20) by $1/\lambda^R_i = c^R_i - \chi n^R_i$ and use (21). For owners, multiply (22) by $1/\lambda^O_i = c^O_i - \chi n^O_i$ and use (23). □

A child's space requirement is priced at the household's *own* shadow value of space, not at the market price. In an unconstrained branch, the full price of a child is $\chi + \kappa q$. For a constrained household it is $\chi + \kappa(q + \zeta^m_i)$: children become more expensive exactly where family-sized space is hard to reach, even with market prices unchanged. Two limits anchor the framework: as $H^m_i \to \infty$ every household pays the unconstrained child price $\chi + \kappa q$, and with $\kappa = 0$ children claim no space and the price reduces to the standard child-goods cost χ. On the financing side, when the collateral component binds, $\partial H^O_i/\partial \tau^p = A_i(x)/((1-\phi)q) > 0$: a higher property tax lowers the asset price at a given user cost and relaxes the down payment, the capitalization channel of Coven et al. (2025).

If no access or menu constraint binds in the relevant tenure branch, the corresponding access value is zero. For example, an unconstrained young owner satisfies

$$\begin{aligned}
c_i^u - \chi n_i^u &= \frac{y_i + A_i(x)}{1+\alpha+\vartheta+k}, \qquad a_i'^u = k\,(c_i^u - \chi n_i^u) + \frac{\ell_{i'} h_i^u}{1+r}, \\
n_i^u &= \frac{\vartheta\,(y_i + A_i(x))}{(1+\alpha+\vartheta+k)(\chi+\kappa q^O)}, \\
h_i^u &= \frac{\alpha\,(y_i + A_i(x))}{(1+\alpha+\vartheta+k)\,q^O} + \kappa \frac{\vartheta\,(y_i + A_i(x))}{(1+\alpha+\vartheta+k)(\chi+\kappa q^O)}.
\end{aligned}$$

The renter analog has the same closed form and is feasible only if $h_i^u \le \bar{h}^R$.

8 Efficiency

At a stationary competitive equilibrium, the price makes aggregate demand equal the fixed stock. Market clearing alone is not enough for efficiency. Conditional on the stationary cross-section and transition margins, efficiency also requires that no feasible reassignment would move marginal space toward households who value it more, net of implementation costs. The test below is compensated: reallocate a unit, hold every household at its prior utility with lump-sum transfers, and ask whether goods are left over. A positive residual is a Pareto improvement under any welfare weights.

Proposition 2 (Planner-equilibrium comparison, local)**.** *Consider an interior stationary competitive equilibrium and a planner with the same preferences, housing stock, tenure size menus, stationary young distribution, old measures, and transition laws held fixed at the instant of the variation. Suppose policy accounts are resource-neutral and the planner has no social birth payoff, so $\nu_i^g = 0$. Conditional on the equilibrium entry measure and tenure assignments, and away from real menu corners, the competitive allocation solves the planner's conditional allocation problem only if no feasible marginal reallocation of existing floorspace has positive goods-equivalent surplus.*

Let $r_i^F \ge 0$ denote the marginal real resource cost, in consumption-good units, of relaxing young owner i's financial implementation constraints enough to permit one more unit of owner floorspace. If the planner can directly reassign existing floorspace and relieve financing without using real resources, then $r_i^F = 0$; if it must operate through costly financing instruments, r_i^F prices those real costs; and if the constraint stands for an enforcement primitive no instrument relaxes, the variation is infeasible, the limit $r_i^F = \infty$. For a young owner whose owner menu is slack and whose financial constraints bind, and for an old incumbent at an interior retained-housing choice, the planner's marginal surplus from increasing the young owner's space by one unit and reducing the old incumbent's retained space by one unit is

$$\left(q^O + \zeta_i^{O,F} - r_i^F\right) - (q - \ell_j) = \zeta_i^{O,F} + \frac{\ell_{i'}}{1+r} + \ell_j - r_i^F.$$

The buyer's anticipated tax is a transfer, like the incumbent's realized one, so it belongs in the recoverable surplus. Thus, in the zero-real-cost finance case, any such pair with $\zeta_i^{O,F} + \ell_{i'}/(1+r) + \ell_j > 0$ prevents the competitive allocation from solving the planner's conditional allocation problem. More generally, a positive young financial access gap is inefficient whenever it can be relaxed at marginal real cost below the value gap relative to the marginal source of space, and a positive retention tax ℓ_j marks an inefficiency whenever the released space has a feasible recipient whose marginal value exceeds the old incumbent's true marginal value.

Conversely, if the financial implementation gaps are zero for all relevant young owners and $\ell_j = 0$ for old incumbents at interior retention choices, then, conditional on entry, tenure assignments, savings, and real menu constraints, the competitive allocation satisfies the planner's conditional housing and fertility first-order conditions for suitable Pareto weights.

Proof. The result is a direct marginal-variation comparison that preserves aggregate floorspace feasibility. Increase household a's occupied floorspace by $\mathrm{d}h$ and reduce household b's occupied floorspace by the same $\mathrm{d}h$, holding entry, tenure assignments, active real menus, fertility, savings, and stationary transition margins fixed at the instant of the variation. The goods-equivalent first-order welfare effect is

$$\left(\frac{u_{h,a}}{u_{c,a}} - \text{marginal real implementation cost for } a\right) - \frac{u_{h,b}}{u_{c,b}}.$$

For a young owner with a slack owner menu and binding financial implementation constraints,

$$\frac{u^Y_{h,i}}{u^Y_{c,i}} = q^O + \zeta_i^{O,F}.$$

If relaxing those constraints has marginal real resource cost r_i^F, the net marginal value of assigning one more unit of space to that owner is $q^O + \zeta_i^{O,F} - r_i^F$, and the anticipated-tax component $\ell_{i'}/(1+r)$ of q^O is a transfer the planner recovers. For an old incumbent at an interior retention choice,

$$\frac{u^O_{h,j}}{u^O_{c,j}} = q - \ell_j.$$

Subtracting gives

$$\zeta_i^{O,F} + \frac{\ell_{i'}}{1+r} + \ell_j - r_i^F.$$

A positive value is a feasible first-order improvement: transfer goods to hold each household at its pre-variation utility, and the residual $\left[\zeta_i^{O,F} + \ell_j - r_i^F\right] \mathrm{d}h$ is available as surplus. Therefore the competitive allocation cannot solve the planner's conditional allocation problem.

If all implementation-relevant financial gaps and retention taxes are zero, the competitive marginal values of all households away from real menu corners coincide with the common market user cost q. Real menu corners are respected by both the competitive economy and the constrained planner. With $\nu_i^g = 0$, the competitive fertility condition then matches the planner's fertility condition evaluated at the same scarcity price, and Pareto weights can be chosen to support the competitive consumption allocation. □

The proposition does not characterize the efficiency of the entry, tenure-assignment, or intergenerational transition margins themselves; those margins are held fixed. It also does not say that every cap is a market failure: real menus remain real constraints, while financial gaps matter only to the extent that an implementable instrument can reach them at a low enough real resource cost.

The misallocation has two sides. A young household that cannot post a larger down payment values the marginal unit of space at $q^O + \zeta_i^{O,F}$, above the market value of that space; the market clears, yet she cannot buy more of it. An old incumbent sitting on a taxable gain keeps the marginal unit while valuing it at $q - \ell_j$, below the market value of that space. Moving one unit from the second household to the first would raise welfare by $\zeta_i^{O,F} + \ell_j - r_i^F$, measured in consumption goods; in the pure-finance case this is $\zeta_i^{O,F} + \ell_j$. Both pieces are computable from data given the preference parameters: the young gap from bundles, via $\zeta_i^{O,F} = \alpha(c_i^O - \chi n_i^O)/(h_i^O - \kappa n_i^O) - q - \ell_{i'}/(1+r)$ when the menu cap is slack—the bundle gap over market rent net of the known tax term— and the old gap from deeds and the tax schedule, via the purchase basis, the current value, the exclusion, and the rate. The old-side force amplifies the misallocation but is not needed for it: the young-access gap is positive whenever a purchase constraint binds with positive shadow value, with or without the gains tax.

Two qualifications matter. First, old occupancy is not inefficient by itself. An old household that keeps a large house because she values it, or because she intends to leave it, is using housing efficiently; those motives are inside u^O and the planner counts them. The inefficiency is only the part of retention supported by the forgiveness at death. Second, not every binding constraint is an inefficiency. The size menus are real features of the housing stock: when a renter is pinned at $\bar{h}^R$, the planner—facing the same stock—is pinned with her, and the multiplier on the menu is the shadow value of a real constraint, not a market failure. Changing the menus requires construction, conversion, or reassignment instruments, whose resource costs belong in Ξ^Π. Down-payment and payment-to-income limits are financing arrangements; their efficiency cost depends on whether feasible instruments can reach them.

A worked example. A two-type stationary economy puts numbers on every object above. Normalize $q = 1$ and set $\rho = 0.25$, so $P = 4$. The young household has $y = 1$, $a = 0.20$, $B_i = 0$, $\alpha = 0.5$, $\vartheta = 0.4$, $\kappa = 0.3$, and the equal-scaling child cost $\chi = \kappa q/\alpha = 0.6$; financing is $\phi = 0.80$ and $\psi = 0.30$; the menus are $\bar{h}^R = 0.15$ and $\bar{h}^O = 1$. The gains tax is $\bar{\ell} = 0.2$ per unit, with the exclusion below the example's sizes, so buyers anticipate it; the period return is $r = 0.35$, so the anticipated-tax wedge is $\bar{\ell}/(1+r) = 0.148$ and the buyer's effective price is $q^O = 1.148$; the discount factor is $\beta = 0.1763$, so with $\gamma = 0.5$ and no bequest motive the savings load is $k = \beta(1+\gamma) = 0.264$. The old incumbent has $a_j = 0.2$, carried housing $H^0 = 0.25$, and tax $\ell = \bar{\ell} = 0.2$. The stock $\bar{H} = 5/12$ is chosen so that $q = 1$ is the market-clearing user cost; entry is deterministic with the inside value above the outside option, and the renewal inflow $\bar{R} = 1 - n^*$ sustains the unit cohort.

The young household buys. Both tenures price space at q, so the branch with the larger cap wins: the owner cap is $H^O = \min\{1, a\rho/((1-\phi)q), \psi y/q\} = \min\{1, 0.25, 0.30\} = 0.25$, the collateral component, against the rental menu 0.15. The cap binds—the unconstrained optimum is $h^u = 0.312$—so $h = 0.25$. The budget is $c + qh + a' = y + a = 1.20$; the savings rule gives $a' = k(c - \chi n) + \bar{\ell}h/(1+r) = 0.163 + 0.037 = 0.20$, so the buyer carries exactly her initial wealth, and the fertility condition gives $n^* = 0.223$, with adult consumption $c - \chi n = 0.616$ and adult space $h - \kappa n = 0.183$. The bundle reveals a gap of $\alpha(c - \chi n)/(h - \kappa n) - q = 0.68$ over market rent: 0.148 of it is her own discounted future tax, and the rest, $\zeta^{O,F} = 0.53$, is financing—she would pay above her effective price for one more unit of space and cannot post the down payment to get it. The incumbent's released units net $q - \ell$, so her liquidation wealth is $a_j + (q - \ell)H^0 = 0.40$ and she keeps $h^O = 1/6$ at the interior solution of $\gamma c_j^O/h_j^O = q - \ell$, valuing retained space at 0.80; her released 1/12 plus the passive stock 1/6 house the young buyer, and the market clears at $\bar{H} = 5/12$. The next old cohort carries the young owner's purchase, 0.25, and her savings return $(1+r)a'$, so the stock recirculates; the incumbent's wealth and tax stand in for her cohort's realized repricing, which the buyer does not anticipate. Figure 1 draws the two sides.

Every object can be read off this one allocation: moving one unit of space from the incumbent to the young household frees $\zeta^{O,F} + \bar{\ell}/(1+r) + \ell = 0.88$ in goods; the cap derivative is $\mathrm{d}n/\mathrm{d}H = 0.30$; and a down-payment grant passes through at $\partial H^O/\partial A_i = \rho/((1-\phi)q) = 1.25$, so a wealth transfer of 0.01 raises the cap by 0.0125 and completed fertility by 0.0038 through access, plus 0.0012 through the budget now that wealth is spendable, 0.0050 in all. The gap, the tax, and the responses are

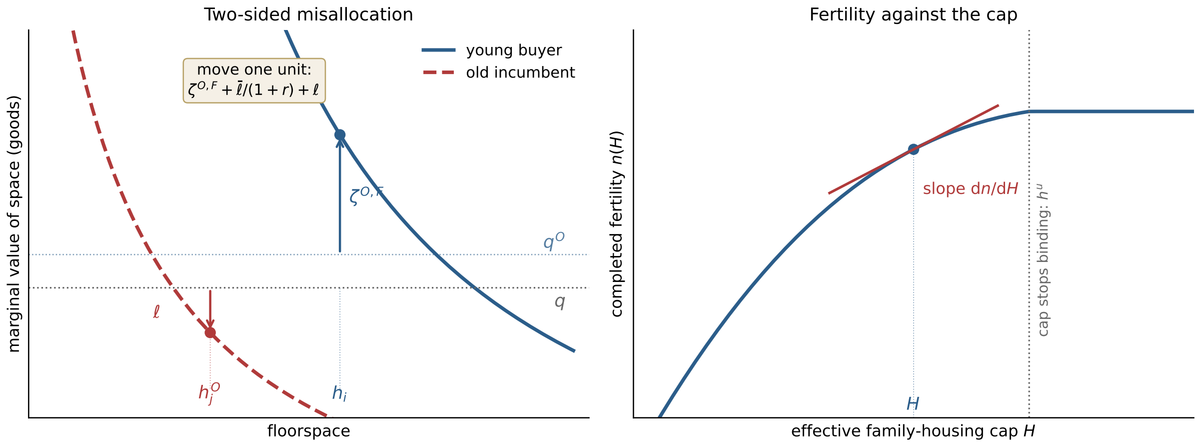

Figure 1: The worked example. Left: marginal value of space. The collateral-capped young owner sits at $h = 0.25$ valuing space at $q^O + \zeta^{O,F} = 1.68$; the incumbent retains $h^O = 1/6$ valuing it at $q - \ell = 0.80$; moving one unit from her to the buyer frees 0.88 in goods. Right: completed fertility against the effective cap. At the equilibrium cap the slope is $\mathrm{d}n/\mathrm{d}H = 0.30$, and the cap stops binding at $h^u = 0.31$.

read off the bundle, the tax schedule, and the financing terms.

9 Fertility and entry

The fertility implication does not require a planner who directly values births. It follows from the private household problem. Children require consumption and housing services, so a constraint on family-capable housing can change fertility even when fertility enters only parental utility.

Fix a tenure branch and suppress the type index. Let H be the effective upper bound on occupied housing in that branch; the user cost is q. For renters, $H = \bar{h}^R$. For young owners,

$$H = \min\left\{\bar{h}^O, \frac{A_i(x)\rho}{(1-\phi)q}, \frac{\psi y_i}{q}\right\}.$$

Holding $(w, \chi, q^m, \alpha, \vartheta, \kappa, k)$ fixed, where $w \equiv y + A_i(x)$ is lifetime resources and q^m is the branch's effective price of space—q for renters and below-exclusion buyers, $q + \bar{\ell}/(1+r)$ for buyers above the exclusion—the branch problem adds savings to the bundle choice. Concentrating the savings rule $a' = k(c - \chi n) + \ell_{i'} h/(1+r)$ into the objective and the budget reduces it to

$$\max_{c,h,n} \{(1+k)\log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta \log n\}$$

up to a type constant, subject to

$$(1+k)(c - \chi n) + \chi n + q^m h = w, \qquad h \leq H, \qquad c - \chi n > 0, \qquad h - \kappa n > 0:$$

the static problem with consumption weight $1 + k$, resources w, and the branch price q^m. Each unit of adult consumption commits $1 + k$ units of resources, one spent now and k carried for old age; each unit of owner space above the exclusion commits its market cost now plus the discounted tax, saved in advance. The unconstrained branch allocation is

$$c^u - \chi n^u = \frac{w}{1+\alpha+\vartheta+k}, \qquad n^u = \frac{\vartheta w}{(1+\alpha+\vartheta+k)(\chi+\kappa q^m)},$$

$$h^u = \frac{\alpha w}{(1+\alpha+\vartheta+k)\, q^m} + \kappa \frac{\vartheta w}{(1+\alpha+\vartheta+k)(\chi+\kappa q^m)}.$$

Suppose $0 < H < h^u$, so the housing cap binds. Then $h(H) = H$, and the constrained fertility choice $n(H)$ is the unique solution to

$$\frac{\vartheta}{n} = \frac{(1+k)\,\chi}{w - q^m H - \chi n} + \frac{\alpha\kappa}{H - \kappa n}.$$

Implicit differentiation gives

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\frac{\alpha\kappa}{(H-\kappa n)^2} - \frac{(1+k)\,\chi q^m}{(w-q^m H-\chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1+k)\,\chi^2}{(w-q^m H-\chi n)^2} + \frac{\alpha\kappa^2}{(H-\kappa n)^2}}.$$

Therefore a relaxation of the binding housing cap raises fertility at H if and only if

$$\alpha\kappa\,(w - q^m H - \chi n)^2 > (1+k)\,\chi q^m\,(H - \kappa n)^2.$$

This condition has a direct Stone–Geary interpretation. At an unconstrained interior branch allocation,

$$\frac{q^m\,(h^u - \kappa n^u)}{c^u - \chi n^u} = \alpha,$$

unchanged by the savings margin and evaluated at the branch price. The child input bundle has the same housing-to-consumption expenditure ratio as the adults' own bundle when

$$\frac{q\kappa}{\chi} = \alpha, \qquad \text{equivalently} \qquad \chi = \frac{\kappa q}{\alpha},$$

the equal-scaling benchmark. With savings, the threshold for the fertility response sits above it: fertility is strictly increasing in every binding cap relaxation if and only if

$$\chi \le (1+k)\,\frac{\kappa q^m}{\alpha},$$

because forced spending on housing now crowds out saving alongside consumption, and only the share $1/(1+k)$ of the budget hit lands on the adult-consumption margin that prices children. A binding cap implies $u_h/u_c = \alpha(c-\chi n)/(H-\kappa n) > q^m$, so the condition above holds strictly under equal scaling whenever the cap binds. Hence, under equal scaling, every relaxation of a strictly binding family-housing cap raises fertility until the cap ceases to bind. If $\chi > (1+k)\kappa q^m/\alpha$, fertility need not rise from every cap relaxation; in particular, $\mathrm{d}n/\mathrm{d}H < 0$ for binding caps sufficiently close to the unconstrained optimum. The reason is that a larger cap raises housing slack, which lowers the crowding cost of children, but it also induces the household to buy more housing at price q, reducing consumption. Fertility rises when the slack effect dominates the consumption-cost effect.

This branch result maps into the model's access margins. A relaxation of the rental size cap raises $H = \bar{h}^R$. If the owner down-payment constraint is the unique binding component of the effective owner cap, then

$$\frac{\partial H}{\partial A_i} = \frac{\rho}{(1-\phi)q} > 0.$$

If the payment-to-income constraint is the unique binding component, then

$$\frac{\partial H}{\partial \psi} = \frac{y_i}{q} > 0.$$

The derivative above signs the fertility response to these primitive cap relaxations locally, holding tenure, prices, income, and child costs fixed.

The competitive-equilibrium/planner comparison is a different statement because prices, consumption, tenure assignments, and entry can all change. In the competitive equilibrium, an entered young household in branch m satisfies

$$\frac{\vartheta\,(c_i^{CE,m} - \chi n_i^{CE,m})}{n_i^{CE,m}} = \chi + \kappa(q^{CE} + \zeta_i^{CE,m}).$$

A fertility-neutral planner has no direct social birth payoff. If the planner allocation for the same type and branch is interior with respect to any remaining access constraint, then

$$\frac{\vartheta\,(c_i^{P,m} - \chi n_i^{P,m})}{n_i^{P,m}} = \chi + \kappa Q^P.$$

Hence

$$n_i^{P,m} > n_i^{CE,m} \quad \iff \quad \frac{c_i^{P,m} - \chi n_i^{P,m}}{c_i^{CE,m} - \chi n_i^{CE,m}} > \frac{\chi + \kappa Q^P}{\chi + \kappa(q^{CE} + \zeta_i^{CE,m})}.$$

Thus a fertility-neutral planner does not mechanically deliver higher fertility merely because the competitive allocation has a positive access value. The branch-level result gives a primitive sufficient condition for a signed fertility response when a policy or planner locally raises a type's effective housing cap holding the branch primitives fixed.

If the planner directly values births through $\mathcal{G}_i(n_i)$, the interior planner condition becomes

$$\frac{\vartheta\,(c_i^{P} - \chi n_i^{P})}{n_i^{P}} + \nu_i^g = \chi + \kappa Q^P.$$

A positive ν_i^g is a separate fertility motive. Holding consumption, the scarcity price of space, and any remaining access constraints fixed, it raises desired fertility. In the full equilibrium, aggregate fertility also depends on prices, consumption, tenure assignments, and entry.

Under the baseline normalization $\bar{n}_i^E = 0$, births generated by the potential-young cohort are

$$N = \bar{M} \int \pi_i^E n_i^Y \,\mathrm{d}G_Y(i).$$

If the outside option has fertility $\bar{n}_i^E$, then cohort births are instead

$$N = \bar{M} \int \left[\pi_i^E n_i^Y + (1 - \pi_i^E)\bar{n}_i^E\right] \mathrm{d}G_Y(i),$$

and every entry-margin term below replaces n_i^Y by $n_i^Y - \bar{n}_i^E$. In the stationary demographic equilibrium, this birth mass and the renewal inflow satisfy $N + \bar{R} = \bar{M}$; the local comparisons below hold $(\bar{M}, G_Y, F_O, F_R, \bar{R})$ fixed and do not by themselves trace the new stationary demographic fixed point after a policy change. Across two allocations,

$$N^P - N^{CE} = \bar{M} \int \left[n_i^P(\pi_i^P - \pi_i^E) + \pi_i^E(n_i^P - n_i^{CE})\right] \mathrm{d}G_Y(i),$$

where n_i^{CE} is fertility in the competitive-equilibrium inside branch and outside-option fertility is normalized to zero. The second term is the intensive fertility margin among the households who would enter in the competitive equilibrium. The first term is the entry margin induced by the change in inside value. If entry is held fixed, the aggregate fertility effect is just

$$N^P - N^{CE} = \bar{M} \int \pi_i^E (n_i^P - n_i^{CE})\, \mathrm{d}G_Y(i).$$

With entry held fixed, aggregate fertility rises only where the planner raises conditional fertility for constrained young types. With entry endogenous, higher inside values can also add entrants.

9.1 A local policy decomposition

Let z be a policy parameter. The local fixed-price effect depends on constrained exposure, pass-through to the binding cap, and the fertility response. For renters, $H_i^R = \bar{h}^R$. For owners,

$$H_i^O = \min\left\{\bar{h}^O, \frac{A_i(x)\rho}{(1-\phi)q}, \frac{\psi y_i}{q}\right\}.$$

Let $m \in \{R, O\}$ denote the active tenure branch and let $\mathcal{E}_m(z)$ be the set of types for whom branch m is uniquely active and $H_i^m(z)$ is the unique binding family-housing cap. Define the policy pass-through to the effective cap by

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m}{\partial z}.$$

Because the cap is a minimum, a policy that improves a slack component does nothing: a subsidy to units below the binding size does not relax a binding down-payment constraint, and a down-payment grant does nothing for a renter pinned by the rental menu. If a policy does not relax the binding component of H_i^m, then $\mathcal{P}_i^m(z) = 0$ for this family-space-cap mechanism. Policies that operate primarily through the price q are a different exercise; they are outside this fixed-price formula, not evaluated by it. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the decomposition isolates the access channel, and an instrument that both relaxes the cap and transfers resources adds the standard income term.

For capped types, the fixed-branch fertility response is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\chi q^m}{(1+k)(c_i^m - \chi n_i^m)^2}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}}.$$

If $\chi \le (1+k)\kappa q^m/\alpha$ and the cap binds strictly, then $\Psi_i^m > 0$. The boundary case $\chi = (1+k)\kappa q^m/\alpha$ is equal scaling at lifetime prices: the child's consumption requirement matches its space requirement when adult consumption is priced at its lifetime cost $1 + k$ and space at the branch price.

Let

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \lambda_i^m \zeta_i^m$$

be the utility value of relaxing the effective cap. Holding prices, tenure choices, the active sets $\mathcal{A}$—the collection of branch assignments and binding cap components—and the stationary demographic state fixed, the local aggregate fertility effect is

$$\left.\frac{\mathrm{d}N}{\mathrm{d}z}\right|_{q,o,\mathcal{A}} = \bar{M} \sum_{m\in\{R,O\}} \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E)\frac{\pi_i^E(1-\pi_i^E)}{\kappa_E}\Theta_i^m\right] \mathcal{P}_i^m(z)\mathrm{d}G_Y(i),$$

where $\bar{n}_i^E = 0$ under the baseline outside-option normalization. The first term is the intensive margin: already-entered constrained households adjust fertility by Ψ_i^m per unit of cap relief, positive under $\chi \le (1+k)\kappa q^m/\alpha$. The second is the entry margin: relaxing a binding cap raises the value of entering the market, and entrants bring their fertility with them—net of the births they would have had outside. Both margins are gated by the same exposure set $\mathcal{E}_m(z)$: a policy aimed at unconstrained households moves nothing.

At the lifetime equal-scaling point, one observable object drives both margins.

Corollary 1 (Equal scaling and access gaps)**.** *Suppose $\chi = (1+k)\kappa q^m/\alpha$, equal scaling at lifetime prices in branch m, and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha}\frac{\zeta_i^m}{c_i^m - \chi n_i^m}\Omega_i^m, \qquad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Under lifetime equal scaling, the access gap ζ_i^m measures both misallocation and fertility exposure. The entry margin runs on the same gap, since $\Theta_i^m = \zeta_i^m/(c_i^m - \chi n_i^m)$. Policies aimed at the largest normalized gaps $\zeta_i^m/(c_i^m - \chi n_i^m)$ raise births the most, holding pass-through, entry, and the weight Ω_i^m fixed.

Proof. At $\chi = (1+k)\kappa q^m/\alpha$, the numerator of Ψ_i^m is

$$\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\kappa (q^m)^2}{\alpha\,(c_i^m - \chi n_i^m)^2} = \frac{\kappa}{\alpha}\left(\frac{\alpha}{h_i^m - \kappa n_i^m} - \frac{q^m}{c_i^m - \chi n_i^m}\right)\left(\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}\right):$$

the $1+k$ in the child cost cancels the $1+k$ in the consumption term of the numerator. At a binding cap, $\alpha(c_i^m - \chi n_i^m)/(h_i^m - \kappa n_i^m) = q^m + \zeta_i^m$, so the first parenthesis equals $\zeta_i^m/(c_i^m - \chi n_i^m)$. Dividing by the positive denominator gives the expression above. □

Thus the fixed-branch effect is exposure to a binding family-space constraint, times pass-through to the effective cap, times the fertility response, plus the induced entry response. The decomposition also holds tenure assignments fixed. With deterministic tenure choice, a policy that changes $W_i^O - W_i^R$ moves the indifference boundary $W_i^O = W_i^R$. If the type distribution has density near that boundary and fertility differs across branches, this tenure-switching margin is first order. The formula above is therefore the fixed-tenure component of the local fertility effect. The next subsection characterizes this margin. A full general-equilibrium policy experiment also changes the price, active sets, old housing choices, fiscal transfers, and possibly the resource cost Ξ^Π.

9.2 Tenure switching at the boundary

In this subsection q^R and q^O are tenure-mode effective user costs. The model now generates its own wedge: a buyer above the exclusion pays $q^O = q + \bar{\ell}/(1+r)$ per unit while a renter pays $q^R = q$; below the exclusion the modes price space identically. The anticipated tax also adds a constant to the owner branch's continuation—a locked-in old household occupies space at $q - \bar{\ell}$, an old-age subsidy worth $\beta\gamma \log\big(q/(q-\bar{\ell})\big)$ in value terms—which shifts tenure choice without moving any bundle. The proposition below carries all three branch differences—effective price, cap, and continuation constant—explicitly; tenure-specific amenities, moving costs, or taste shifters still require a separate calculation.

Tenure switching is not a price or general-equilibrium effect. It is already present in the fixed-price model because tenure is chosen by

$$W_i^Y = \max\{W_i^R, W_i^O\}.$$

A policy that raises the owner value relative to the renter value moves households near the boundary $W_i^O = W_i^R$ from renting to owning, and because tenure is a discrete choice, the switchers' bundles—and possibly their fertility—jump. Whether they jump depends on a single feature of the environment: whether the two tenures price a unit of space differently. Below the exclusion they price it identically and the jump is exactly zero: there, the local policy formula of the previous subsection is complete. Above the exclusion the anticipated tax makes owning the expensive way to occupy space, and the jump is signed accordingly; further wedges—a rental premium for large units as in the smooth menu formulation, landlord operating costs—move the gap the same way or against it.

The branch problems include savings. Concentrating the rule as above turns each branch into the static problem with consumption weight $1 + k$ and branch price q^m, plus a branch constant: with common old preferences and interior old choices, the owner branch's continuation exceeds the renter's by

$$\Delta = \beta\gamma \log \frac{q}{q - \bar{\ell}} > 0$$

above the exclusion—the old-age subsidy of occupying retained space at $q - \bar{\ell}$—and the constants coincide below it. The constant moves tenure choice without moving any bundle, so the proposition carries it explicitly: write K^A and K^B for the branch constants and $\Delta = K^B - K^A \geq 0$.

Fix prices and suppress the type index. Let A denote the cheaper branch and B the more expensive branch, with tenure-mode user costs

$$q^A < q^B.$$

Let (c_m, h_m, n_m) denote the branch-m optimum, $m \in \{A, B\}$. Let ζ^m be the goods-equivalent value of relaxing the effective family-housing cap in branch m.

Proposition 3 (Fertility at a tenure switch)**.** *Suppose $q^A < q^B$, the branch continuations differ by $\Delta = K^B - K^A \geq 0$, and the household is indifferent, $W^A = W^B$, so the dear branch runs a static-value deficit of exactly Δ. Then:*

(i) if the price gap dominates the constant,

$$\alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A} > \Delta,$$

the cheaper branch's effective cap binds with positive value, $h_A = H^A$ and $\zeta^A > 0$;

(ii) if moreover the wedge dominates the constant in resources,

$$(q^B - q^A)\, h_B > \Delta\, (c_A - \chi n_A),$$

the switcher upsizes and trades consumption for space: $h_B > h_A$ and $c_B - \chi n_B < c_A - \chi n_A$;

(iii) if in addition $\chi \leq (1 + k)\kappa q^A/\alpha$, then $n_B > n_A$ and $h_B - \kappa n_B > h_A - \kappa n_A$ whenever the within-set force exceeds the level shift, the inequality displayed in the proof; with $\Delta = 0$ the conditions in (i) and (ii) hold automatically, the level shift vanishes, and all conclusions reduce to the earlier ones.

Proof. Write $W^m = \tilde{W}^m + K^m$ up to common terms, where $\tilde{W}^m$ is the concentrated static value at price q^m and cap H^m; indifference is $\tilde{W}^A - \tilde{W}^B = \Delta$.

(i) If $\zeta^A = 0$, branch A attains the unconstrained static value, and the closed forms give

$$\tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B) = \alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A}.$$

Since $\tilde{W}^B \le \tilde{W}_{\text{unc}}(q^B)$, the deficit would satisfy $\Delta \ge \tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B)$, contradicting price dominance. Thus $h_A = H^A$ and $\zeta^A > 0$.

(ii) Suppose $h_B \le h_A$. The branch-B bundle is then feasible in branch A and costs $(q^B - q^A)h_B$ less. The static value is concave in resources with marginal utility $1/(c_A - \chi n_A)$ at w, so

$$\Delta = \tilde{W}^A - \tilde{W}^B \ge \frac{(q^B - q^A)\, h_B}{c_A - \chi n_A},$$

contradicting wedge dominance. Thus $h_B > h_A$.

Every branch optimum satisfies

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha\kappa}{h - \kappa n}.$$

Define

$$X = \frac{\chi}{c - \chi n}, \qquad Y = \frac{\alpha\kappa}{h - \kappa n}.$$

X is large when adult consumption is squeezed, Y when adult space is squeezed.[7] Then $n = \vartheta/(X+Y)$, and branch utility can be written as $K^m - \Phi(X,Y)$ up to a common type constant, where

$$\Phi(X,Y) = (1+k)\log X + \alpha \log Y + \vartheta \log(X+Y).$$

Indifference places the two optima on level sets that differ by the constant:

$$\Phi(X_B, Y_B) = \Phi(X_A, Y_A) + \Delta.$$

Occupied housing pins down the position along a level set: at any branch optimum,

$$h = \kappa \left(\frac{\alpha}{Y} + \frac{\vartheta}{X+Y} \right) = \kappa\, \Phi_Y(X,Y).$$

Let $T = X + Y$. Along a level set of Φ,

$$\left.\frac{\mathrm{d}\Phi_Y}{\mathrm{d}X}\right|_{\Phi} = \frac{\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X}{\Phi_Y},$$

and

$$\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X = \frac{(1+k)\,\alpha}{XY^2} + \frac{(1+k)\,\vartheta}{XT^2} + \frac{\alpha\vartheta X}{Y^2T^2} > 0,$$

so on every level set Φ_Y is strictly increasing in X; and at fixed X, moving to a higher level set raises Y and, since $\Phi_{YY} < 0$, lowers Φ_Y. Hence $h_B > h_A$ forces $X_B > X_A$, and $c_B - \chi n_B = \chi/X_B < \chi/X_A = c_A - \chi n_A$.

[7] Capital X and Y are local to this proof and unrelated to the bequest share x and income y_i.

(iii) Write $T(X;\varphi)$ for $X+Y$ along the level set $\Phi=\varphi$, and decompose along the path that runs from (X_A,Y_A) along Φ_A to X_B, then up to $\Phi_A+\Delta$ at fixed X_B:

$$T_B - T_A = \underbrace{\int_{\Phi_A}^{\Phi_A+\Delta} \frac{\mathrm{d}\varphi}{\Phi_Y(X_B,\cdot)}}_{\text{level shift } \geq 0} - \underbrace{\int_{X_A}^{X_B} \frac{\Phi_X-\Phi_Y}{\Phi_Y}\bigg|_{\Phi_A} \mathrm{d}X}_{\text{within-set force}}.$$

The within-set integrand is positive exactly where $(1+k)Y>\alpha X$, the region above the ray $Y=\alpha X/(1+k)$, which crosses each level set once: the level set is a decreasing graph, while along the ray $\Phi=(1+k+\alpha+\vartheta)\log X+\text{constant}$ is strictly increasing in X. At a branch optimum,

$$\frac{\alpha\,(c_m-\chi n_m)}{h_m-\kappa n_m}=q^m+\zeta^m, \qquad \zeta^m\geq 0, \qquad \text{so} \qquad \frac{Y_m}{X_m}=\frac{\kappa(q^m+\zeta^m)}{\chi},$$

and under $\chi\leq(1+k)\kappa q^A/\alpha$ both optima lie strictly in the region: branch A because $\zeta^A>0$, branch B because $q^B>q^A$. Then $n_B=\vartheta/T_B>\vartheta/T_A=n_A$ if and only if the within-set force exceeds the level shift. When it does, $T_B<T_A$ together with $X_B>X_A$ gives $Y_B<Y_A$, hence $h_B-\kappa n_B=\alpha\kappa/Y_B>h_A-\kappa n_A$. With $\Delta=0$ the level shift vanishes, the path stays in the region between the two optima, and the inequality always holds. □

At indifference there are now two reasons to pay more per unit of space: to get more space, and to buy the dear branch's better old age. The conditions price that second temptation: the wedge must dominate the subsidy in resources for the switcher to upsize, and the fertility jump nets the within-branch space force against the subsidy's level shift. At the example's parameters the conditions hold with room to spare—price dominance by a factor of four—and the boundary household upsizes from 0.15 to 0.20 and adds 0.007 children.

The rent-own cases now read off directly, with the effective user costs $q^R=q$ and $q^O=q+\ell_{i'}/(1+r)$. *Below the exclusion ($q^O=q^R=q$):* the modes differ only through their effective caps, and an indifferent household has the same allocation in both—either both caps are slack, or the caps coincide at the binding allocation. Then $n_O=n_R$, there is no discrete jump at the boundary, and the fixed-tenure formula of the previous subsection is the complete fixed-price answer. *Above the exclusion ($q^O>q^R$):* ownership is the expensive family-space mode, and an indifferent switcher into ownership upsizes and has $n_O>n_R$; this is the model's own configuration for family-sized purchases. *Renting dearer ($q^O<q^R$, for example a large-unit rental premium):* the inequality reverses; the marginal switcher into ownership moves to a cheaper, smaller unit, and $n_R>n_O$ under the analogous condition at the lower owner user cost. A cost wedge between the tenures is what makes tenure switching a first-order fertility-jump margin.

The switching formula is fixed-price accounting, not a welfare result or a general-equilibrium comparative static. Let types live in a smooth finite-dimensional type space with continuous density $f(i)$, and suppose there are no mass points at the tenure boundary. Let

$$\Delta_i(z)=W_i^O(z)-W_i^R(z), \qquad \mathcal{I}(z)=\{i:\Delta_i(z)=0\},$$

and assume $\nabla_i\Delta_i(z)\neq 0$ on $\mathcal{I}(z)$, with unique branch optima and kinks in active constraints having zero measure. The switching component of the fixed-price fertility effect is

$$\frac{\mathrm{d}N^{\text{switch}}}{\mathrm{d}z}\bigg|_{q,\mathcal{A}}=\bar{M}\int_{\mathcal{I}(z)}\pi_i^E\left(n_i^O-n_i^R\right)\frac{\partial_z\Delta_i(z)}{\|\nabla_i\Delta_i(z)\|}f(i)\,\mathrm{d}S(i),$$

where $\mathrm{d}S(i)$ is surface measure on the indifference boundary. For cap-shifting policies,

$$\partial_z\Delta_i(z)=\Theta_i^O\mathcal{P}_i^O(z)-\Theta_i^R\mathcal{P}_i^R(z), \qquad \Theta_i^m=\lambda_i^m\zeta_i^m.$$

Corollary 2 (Owner-access policies)**.** *Consider an owner-access policy with* $\mathcal{P}_i^O(z) \geq 0$ *and* $\mathcal{P}_i^R(z) = 0$*. Suppose the conditions of Proposition 3, including part (iii)'s inequality, hold for almost every type on* $\mathcal{I}(z)$*. If* $q^O > q^R$*, the switching component is nonnegative, so the fixed-tenure owner-access formula is a lower bound for this fixed-price fertility channel. If* $q^O = q^R$ *(the baseline), the switching component is zero. If* $q^O < q^R$*, the switching component is nonpositive.*

Corollary 3 (Rental-menu policies)**.** *Consider a rental-menu policy with* $\mathcal{P}_i^R(z) \geq 0$ *and* $\mathcal{P}_i^O(z) = 0$*. Suppose the conditions of Proposition 3, including part (iii)'s inequality, hold for almost every type on* $\mathcal{I}(z)$*. If* $q^O > q^R$*, the switching component is nonpositive: relaxing the rental cap raises fertility for constrained renters who remain renters, but it can also pull marginal owners back into renting. If* $q^O = q^R$ *(the baseline), the switching component is zero. If* $q^O < q^R$*, the switching component is nonnegative.*

References

Barro, Robert J., and Gary S. Becker. 1988. "A Reformulation of the Economic Theory of Fertility." *Quarterly Journal of Economics* 103(1): 1–25.

Barro, Robert J., and Gary S. Becker. 1989. "Fertility Choice in a Model of Economic Growth." *Econometrica* 57(2): 481–501.

Coven, Joshua, Sebastian Golder, Arpit Gupta, and Abdoulaye Ndiaye. 2025. "Property Taxes and Housing Allocation Under Financial Constraints." Working paper, June 2025. Earlier version: CESifo Working Paper No. 11203, 2024.

Diamond, Peter A. 1965. "National Debt in a Neoclassical Growth Model." *American Economic Review* 55(5): 1126–1150.

Fonseca, Julia, Lu Liu, and Pierre Mabille. 2024. "Unlocking Mortgage Lock-In: Evidence from a Spatial Housing Ladder Model." Working paper.

Galor, Oded, and David N. Weil. 1996. "The Gender Gap, Fertility, and Growth." *American Economic Review* 86(3): 374–387.

Gervais, Martin. 2002. "Housing Taxation and Capital Accumulation." *Journal of Monetary Economics* 49(7): 1461–1489.

Sommer, Kamila, and Paul Sullivan. 2018. "Implications of US Tax Policy for House Prices, Rents, and Homeownership." *American Economic Review* 108(2): 241–274.